

Test 14: Spectral rigidity ??(L)

Verdict: PASS | Generated: 2026-09-02 14:02:56 | Suite: AB-Cloud v19 (Julia 1.12.0)

What this test verifies

??(L) long-range rigidity vs GUE universal prediction.

METHOD: ??(L) long-range rigidity vs GUE universal prediction.

Result

Test 14 [HARDCORE pass 2]: 3 sub-checks, 0 failed ? PASS

‘plots/plot_01‘ (.png 600dpi / .svg / .pdf)

Plots

- ‘plots/animation.gif‘ ? animated reveal of every plot of this test

Reproduce with: julia ab_cloud_v19.jl --test 14 --no-two-pass

Independent verification guide

Every computation is logged in logs/computation_log.txt (timestamps + deltas); raw console output in logs/stdout_capture.txt.

[14:02:56.160] [+891.443s] calc: ??(L=2.000) = 0.093421 (n_pos=50000, n_windows=500, n_devs=500)

[14:02:56.181] [+891.463s] calc: ??(L=4.536) = 0.128616 (n_pos=50000, n_windows=500, n_devs=500)

FULL COMPUTATION LOG

[14:02:56.201] [+891.483s] calc: ??(L=10.287) = 0.161422 (n_pos=50000, n_windows=500, n_devs=500)

[14:02:56.219] [+891.501s] calc: ??(L=23.331) = 0.173069 (n_pos=50000, n_windows=500, n_devs=500)

[14:02:56.236] [+891.518s] calc: ??(L=52.915) = 0.175808 (n_pos=50000, n_windows=500, n_devs=500)

[14:02:56.254] [+891.536s] calc: ??(L=120.010) = 0.176269 (n_pos=50000, n_windows=500, n_devs=500)

[14:02:56.273] [+891.555s] calc: ??(L=272.178) = 0.175052 (n_pos=50000, n_windows=500, n_devs=500)

[14:02:56.291] [+891.573s] calc: ??(L=617.292) = 0.176106 (n_pos=50000, n_windows=500, n_devs=500)

[14:02:56.310] [+891.592s] calc: ??(L=1400.000) = 0.175848 (n_pos=50000, n_windows=500, n_devs=500)

[14:02:56.324] [+891.606s] calc: ??(L=23.331) = 0.173347 (n_pos=50000, n_windows=166, n_devs=166)

[14:02:56.330] [+891.612s] calc: ??(L=23.331) = 0.173506 (n_pos=50000, n_windows=166, n_devs=166)

[14:02:56.336] [+891.618s] calc: ??(L=23.331) = 0.173829 (n_pos=50000, n_windows=166, n_devs=166)

??

TEST 14 (secondary, HARDCORE): Objection 2 (extended): Advanced RMT Diagnostics ? ?? audit

??

CONSOLE CAPTURE

L ??_data Poisson=L/15 data/Pois

2.0 0.0934 0.1333 0.7007

4.5 0.1286 0.3024 0.4253

10.3	0.1614	0.6858	0.2354
23.3	0.1731	1.5554	0.1113
52.9	0.1758	3.5277	0.0498
120.0	0.1763	8.0006	0.0220
272.2	0.1751	18.1452	0.0096
617.3	0.1761	41.1528	0.0043
1400.0	0.1758	93.3333	0.0019

G2 monotone : 0 dips >30% across 8 grid steps (limit 1)

G3 seed noise: ??(L=23.3) over 3 seeds: 0.1733/0.1735/0.1738, spread=0.3% (limit 35%)

??

G1 beat Poisson PASS ? 9/9 L-values

G2 monotone PASS ? 0 dips

G3 seed noise PASS ? spread = 0.3%

Test 14 [HARDCORE pass 2]: 3 sub-checks, 0 failed ? PASS